

CS 350: Algorithm Design & Analysis

Homework Set 4 — Due: Friday, Oct 2, 11:59 PM

Instructions

Submit your solutions as a single PDF via Gradescope. Show all work for full credit. Collaboration is allowed, but you must write your own solutions and cite collaborators below.

Problem 1 (20 pts) — Recurrence Relations

Solve the following recurrence using the Master Theorem, or explain why the Master Theorem does not apply. Show which case applies and justify your answer.

$$T(n) = 4T(n/2) + n^2 \log n$$

Problem 2 (25 pts) — Divide and Conquer

Given an unsorted array of n integers, design a divide-and-conquer algorithm that finds both the maximum and minimum elements using at most $3n/2 - 2$ comparisons. Provide pseudocode and prove the comparison bound using induction.

Problem 3 (25 pts) — Graph Traversal

Given a directed acyclic graph (DAG) representing course prerequisites, write an algorithm that outputs a valid course ordering (topological sort). Analyze the time and space complexity in terms of V (vertices) and E (edges).

Problem 4 (30 pts) — Dynamic Programming

You are given a set of n jobs, each with a start time, end time, and profit. Design a dynamic programming algorithm to select a subset of non-overlapping jobs that maximizes total profit (Weighted Job Scheduling). Give the recurrence, pseudocode, and overall time complexity.

Collaborators (if any): _____ | Estimated time spent: _____ hours